Predictive Insight Engine

Supervised Learning — Complete Theory Notes

Based on the project instructions: Regression, Model Evaluation, Polynomial Regression, Gradient Descent, Bias–Variance, Overfitting & Underfitting

Project Objective

The project focuses on understanding, implementing and evaluating supervised learning algorithms, especially regression techniques and model performance analysis. The main goal is to predict house prices from features such as area, bedrooms, bathrooms, location score and property age.

Dataset Features

- House Area (sq.ft)
- Number of Bedrooms
- Number of Bathrooms
- Location Score
- Age of Property
- House Price — Target Variable

1. Supervised Learning Algorithms

Supervised Learning is a type of Machine Learning in which a model learns from **labeled data**. The input features (X) and correct output/target (y) are already known during training. The model learns the relationship between X and y and then predicts the output for new data.
Example: Using house area, bedrooms and location score to predict house price.

2. Regression vs Classification

Regression predicts a continuous numerical value, such as house price, salary or temperature.
Classification predicts a category/class, such as spam/not spam or pass/fail.
In this project, house-price prediction is a **regression** problem because price is numeric and continuous.

3. Simple Linear Regression

Simple Linear Regression predicts a target using **one independent variable**. The basic equation is **y =** β■ **+** β■**x**, where β■ is the intercept and β■ is the slope. The model finds the best straight line that represents the relationship between x and y.
Example: House Area → House Price.

4. Assumptions of Linear Regression

Linear regression works best when important assumptions are reasonably satisfied:
• **Linearity:** Relationship between input and target should be approximately linear.
• **Independence:** Observations/errors should be independent.
• **Homoscedasticity:** Error variance should remain roughly constant.
• **Normality of residuals:** Residuals should be approximately normally distributed, especially for statistical inference.
• **No severe multicollinearity:** Predictors should not be extremely correlated with each other.
These can be checked using scatter plots, residual plots, histograms and correlation analysis.

5. Bias–Variance Trade-Off

Bias is error caused by an overly simple model and strong assumptions. **Variance** is error caused by a model being too sensitive to the training data. A good model balances both.
High bias usually leads to underfitting; high variance usually leads to overfitting. The goal is to find a model complexity that gives good generalization.

6. Overfitting and Underfitting

Overfitting: The model learns training data too closely, including noise, so training performance is very good but test performance becomes poor. A high-degree polynomial can overfit.
Underfitting: The model is too simple to capture the real pattern, so both training and test performance are poor. A straight line may underfit a strongly curved relationship.

7. Independent and Dependent Variables

Independent variables (features/predictors): House Area, Bedrooms, Bathrooms, Location Score and Age of Property.
Dependent variable (target): House Price. The model uses independent variables to predict the dependent variable.

8. Training and Testing Data

The dataset is divided into a **training set** and a **testing set**. The training data is used to learn model parameters, while the testing data is kept separate to evaluate how well the trained model performs on unseen data.
A common approach is an 80:20 or 70:30 split, depending on the project requirement.

9. Multiple Linear Regression

Multiple Linear Regression predicts a target using **two or more independent variables**. A simplified equation is **y = β■ + β■x■ + β■x■ + ... + β■x■**.
For this project, house price can be predicted using area, bedrooms, bathrooms, location score and age together. Performance may improve because more useful information is available, but irrelevant or highly correlated features can also hurt generalization.

10. Polynomial Regression

Polynomial Regression extends linear regression by adding powers of the input, such as x^2 or x^3. For example: **y = β■ + β■x + β■x²**.
It can model curved relationships. Increasing the degree makes the model more flexible, but a very high degree can cause overfitting.

11. Gradient Descent

Gradient Descent is an optimization algorithm used to minimize a loss/cost function. It repeatedly updates model parameters in the direction that reduces error.
Basic idea: calculate the gradient → choose a learning rate → update parameters → repeat until the cost becomes sufficiently small or convergence is reached.
Learning rate: controls the size of each update. Too large can overshoot; too small can make training slow.

12. Batch Gradient Descent

Batch Gradient Descent calculates the gradient using the **entire training dataset** before making one parameter update. It usually gives a stable direction toward the minimum, but each update can be computationally expensive for large datasets.

13. Stochastic Gradient Descent (SGD)

SGD updates model parameters using **one training example at a time**. It can be much faster for large datasets and may start improving quickly, but its path toward the minimum is noisy and less stable than batch gradient descent.

14. Mini-Batch Gradient Descent

Mini-Batch Gradient Descent divides the training data into small batches and updates parameters after each batch. It combines advantages of batch and stochastic approaches: more efficient than full-batch training and generally more stable than pure SGD.

15. Convergence Behavior and Training Time

Convergence means the optimization process approaches a point where further updates produce little improvement in the cost. Batch GD tends to have smooth updates; SGD is noisier; Mini-Batch usually

provides a practical balance. Training time depends on dataset size, batch size, number of iterations/epochs and implementation.

16. Model Evaluation Metrics

The project requires MSE, MAE, RMSE, R^2 and Adjusted R^2. These metrics help compare predicted house prices with actual prices and understand model quality.

17. Mean Squared Error (MSE)

MSE is the average of squared prediction errors:
MSE = average((y − ■)²)
Because errors are squared, large errors receive more penalty. Lower MSE generally indicates better prediction performance.

18. Mean Absolute Error (MAE)

MAE is the average absolute difference between actual and predicted values:
MAE = average(|y − ■|)
It is easy to interpret because it is in the same unit as the target. Lower MAE is better.

19. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE:
RMSE = √MSE
It is in the same unit as the target and penalizes large errors more strongly than MAE. Lower RMSE is better.

20. R^2 Score

R^2 (coefficient of determination) measures how much of the variation in the target is explained by the model. A value closer to 1 generally indicates stronger explanatory/predictive fit, while 0 means the model is no better than predicting the target mean in the standard formulation. It can also be negative on unseen data.

21. Adjusted R^2

Adjusted R^2 modifies R^2 by considering the number of predictors and sample size. It can decrease when unnecessary features are added. Therefore, it is useful when comparing models with different numbers of predictors.

22. Residuals and Model Diagnostics

A **residual** is the difference between an actual value and its prediction: **residual = y** − ■. Residual plots help detect non-linearity, unequal error variance, outliers and other patterns that suggest model assumptions may not hold.

23. Bias and Variance Across Models

Simple Linear Regression is relatively simple and may have higher bias but lower variance. Multiple Linear Regression can capture more information. Polynomial Regression is more flexible: as degree increases, bias can decrease while variance can increase. The best model is not necessarily the most complex; it should generalize well to unseen data.

24. Model Complexity and Prediction Error

Increasing model complexity can initially reduce prediction error because the model captures more structure. After a suitable point, excessive complexity can fit noise and increase test error. This is the central idea behind the bias–variance trade-off.

25. Choosing the Best Model

The best model should be selected using **test/validation performance**, not training performance alone. Compare metrics such as MAE, RMSE and R^2, inspect residuals, and look for evidence of overfitting or underfitting. A model with a good balance of accuracy, simplicity and generalization is preferable.

26. Practical Business Interpretation

For a real-estate analytics company, the model can support approximate house-price estimation from property characteristics. A useful business model should not only have good numerical metrics but also provide stable predictions and understandable relationships between features and price.

Quick Viva Revision

Remember these one-line answers:

- Supervised Learning → learns from labeled input-output data.
- Regression → predicts continuous numerical values.
- Classification → predicts categories/classes.
- Simple Linear Regression → one feature + straight-line relationship.
- Multiple Linear Regression → multiple features used together.
- Polynomial Regression → adds powers of features to model curves.
- Overfitting → great training performance, poor generalization.
- Underfitting → model too simple; poor training and test performance.
- Bias → error from excessive simplicity.
- Variance → sensitivity to training data.
- Gradient Descent → minimizes a cost function by parameter updates.
- Batch GD → update using all training examples.
- SGD → update using one example at a time.
- Mini-Batch GD → update using a small batch.
- MSE → squared-error average.
- MAE → absolute-error average.
- RMSE → square root of MSE.
- R^2 → proportion of target variation explained by the model.
- Adjusted R^2 → R^2 adjusted for number of predictors.
- Residual → actual value minus predicted value.

Project Deliverables from the Instructions

- Jupyter Notebook with practical implementation and screenshots.
- Clearly labeled charts with short interpretations.
- GitHub repository containing source code and documentation.
- Theory concepts document in PDF.
- README.md and video link as required.

These notes are prepared directly from the project instruction topics, expanded into study-friendly theory. The project specifically asks for theory concepts including supervised learning, regression/classification, linear regression assumptions, bias–variance, and overfitting/underfitting, followed by regression models, evaluation metrics, gradient descent variants and diagnostics.